Yatin Singh Jakhar

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EDUCATION

Manav Rachna University *Aug 2024 – Present*
B.Tech in Electronics and Communication Engineering (Very Large Scale Integration)

Eicher School, Faridabad *Jun 2023 – May 2024*
Senior Secondary – 12th (CBSE)

TECHNICAL SKILLS

Hardware & Hardware Description Languages: Verilog, RTL Design, Digital Logic Design, Embedded Systems, Circuit Analysis

Tools & Platforms: EDA Playground, ModelSim

INTERNSHIPS

Pritam Singh Memorial (PRISM) Conference Intern
(Manav Rachna International Institute of Research and Studies) *September 2026 – Present*

- Coordinated academic and corporate logistics: Assisted in organizing a large-scale national management conference, managing delegate communications, registration desks, and scheduling across multi-day events.
- Streamlined event operations: Collaborated with cross-functional committees to manage venue arrangements, technical setups, and hospitality for corporate leaders and academic faculty.
- Supported content and proceedings management: Assisted in compiling research paper abstracts, session summaries, and documentation for conference deliverables and post-event reports.

PROJECTS

6 DOF Robotic Arm Using Joystick Control

- Engineered and led a multi-disciplinary team of 5 as Project Lead to design and build a 6 Degrees of Freedom (DOF) robotic arm powered by an Arduino Uno.
- Integrated mixed-power servo architectures, combining high-torque RC digital servos (35kg) with external power supplies for major joints and SG90 servos for precision wrist and gripper movements.
- Developed custom embedded C++ (Arduino) software featuring joystick-to-servo mapping, deadzone filtering, step-size smoothing, and instant position reset functionality.

4-Bit Palindrome Detector Using FSM & Verilog HDL

- Designed a sequential digital logic circuit using a Finite State Machine (FSM) in Verilog HDL to verify 4-bit serial binary sequences for palindromes.
- Implemented RTL state transitions across five distinct clock cycles to store and compare bit pairs ($B_3 = B_0$ and $B_2 = B_1$)
- Validated the design successfully through rigorous test-bench simulations covering both valid palindrome and non-palindrome edge cases.

TRAININGS & CERTIFICATIONS

- Verilog | Truechip Solutions PVT. LTD. (June 2026)
- Entrepreneurship Essentials | NPTEL, Govt. of India (Jan-Apr 2026)
- VHDL (Basics)| Truechip Solutions PVT. LTD. (June 2025)

EXTRACURRICULAR ACTIVITIES & LEADERSHIP

- Clan Coordinator, Student Welfare: Spearheaded the planning and execution of multiple large-scale student engagement events, managing logistics and team coordination.
- Technical Team Member, Digital Nexus: Organized and facilitated technical workshops and coding events, driving participation and hands-on learning among peers.
- Performer & Representative, Rudra Dance Society: Represented the institution in premier inter-college dance competitions, including a high-profile showcase at the Constitution Club of India..